

Ex No: 1c

IMPLEMENTATION OF MINIMAX algorithm

Aim:

To implement MINIMAX algorithm.

Procedure:

Define player representation constants (PLAYER_X, PLAYER_O, EMPTY)
for the 3×3 Tic-Tac-Toe board.

Create the evaluate function to check for win conditions across rows, columns,
and diagonals, returning the winning player's score or 0

Implement isMovesLeft to check whether any empty spots remain on the
board.

Implement the minimax recursive function to evaluate all possible board states
and return the maximum score for PLAYER_X or minimum score for
PLAYER_O

Define findBestMove to iterate through available empty positions, invoke
minimax, and select the optimal board coordinate for PLAYER_X

Initialize an example board state, display it, calculate the best move using
findBestMove, and print the updated board.

Program:

Constants for players

PLAYER_X = 1

PLAYER_O = -1

EMPTY = 0

Evaluate the board

def evaluate(board):

for row in range(3):

if board[row][0] == board[row][1] == board[row][2] != EMPTY:

return board[row][0]

for col in range(3):

if board[0][col] == board[1][col] == board[2][col] != EMPTY:

return board[0][col]

if board[0][0] == board[1][1] == board[2][2] != EMPTY:

return board[0][0]

if board[0][2] == board[1][1] == board[2][0] != EMPTY:

return board[0][2]

return 0

Check if moves are left

def isMovesLeft(board):

for row in range(3):

for col in range(3):

if board[row][col] == EMPTY:

return True

```
return False
```

```
# Minimax function
```

```
def minimax(board, isMax):
```

```
    score = evaluate(board)
```

```
    if score == PLAYER_X: return score
```

```
    if score == PLAYER_O: return score
```

```
    if not isMovesLeft(board): return 0
```

```
    if isMax:
```

```
        best = -float('inf')
```

```
        for row in range(3):
```

```
            for col in range(3):
```

```
                if board[row][col] == EMPTY:
```

```
                    board[row][col] = PLAYER_X
```

```
                    best = max(best, minimax(board, not isMax))
```

```
                    board[row][col] = EMPTY
```

```
    return best
```

```
else:
```

```
    best = float('inf')
```

```
    for row in range(3):
```

```
        for col in range(3):
```

```
            if board[row][col] == EMPTY:
```

```

board[row][col] = PLAYER_O
best = min(best, minimax(board, not isMax))
board[row][col] = EMPTY

return best

# Find the best move for PLAYER_X
def findBestMove(board):

    bestVal = -float('inf')
    bestMove = (-1, -1)
    for row in range(3):

        for col in range(3):

            if board[row][col] == EMPTY:

                board[row][col] = PLAYER_X
                moveVal = minimax(board, False)
                board[row][col] = EMPTY
                if moveVal > bestVal:

                    bestMove = (row, col)
                    bestVal = moveVal

    return bestMove

# Print the board
def printBoard(board):

    for row in board:

        print(" ".join(["X" if x == PLAYER_X else "O" if x == PLAYER_O else "." for x in

            row]))

```

```
# Example game
board = [

    [PLAYER_X, PLAYER_O, PLAYER_X],
    [PLAYER_O, PLAYER_X, EMPTY],
    [EMPTY, PLAYER_O, PLAYER_X]

]

print("Current Board:")
printBoard(board)

move = findBestMove(board)
print(f"Best Move: {move}")
board[move[0]][move[1]] = PLAYER_X
print("\nBoard after best move:")
printBoard(board)
```

Output:

Current Board:

X O X

O X .

. O X

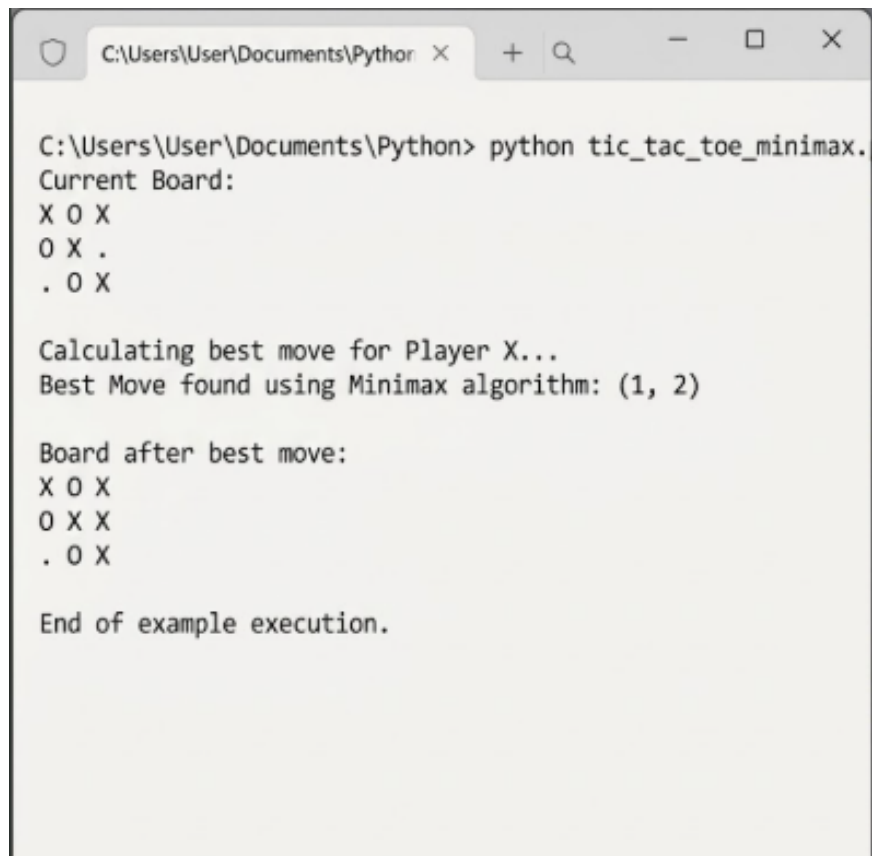
Best Move: (2, 0)

Board after best move:

X O X

O X .

X O X



```
C:\Users\User\Documents\Python> python tic_tac_toe_minimax.  
Current Board:  
X O X  
O X .  
. O X  
  
Calculating best move for Player X...  
Best Move found using Minimax algorithm: (1, 2)  
  
Board after best move:  
X O X  
O X X  
. O X  
  
End of example execution.
```